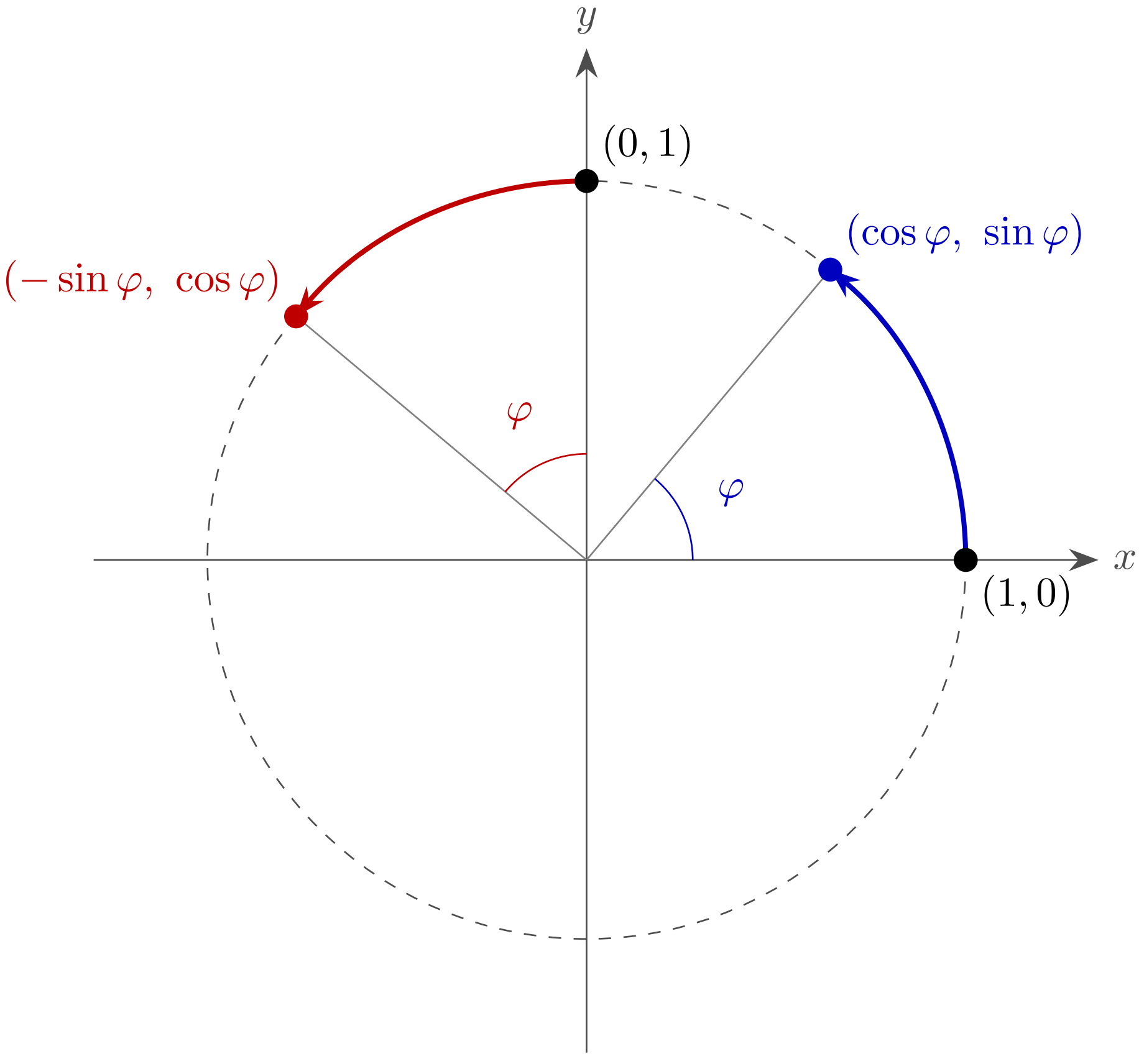
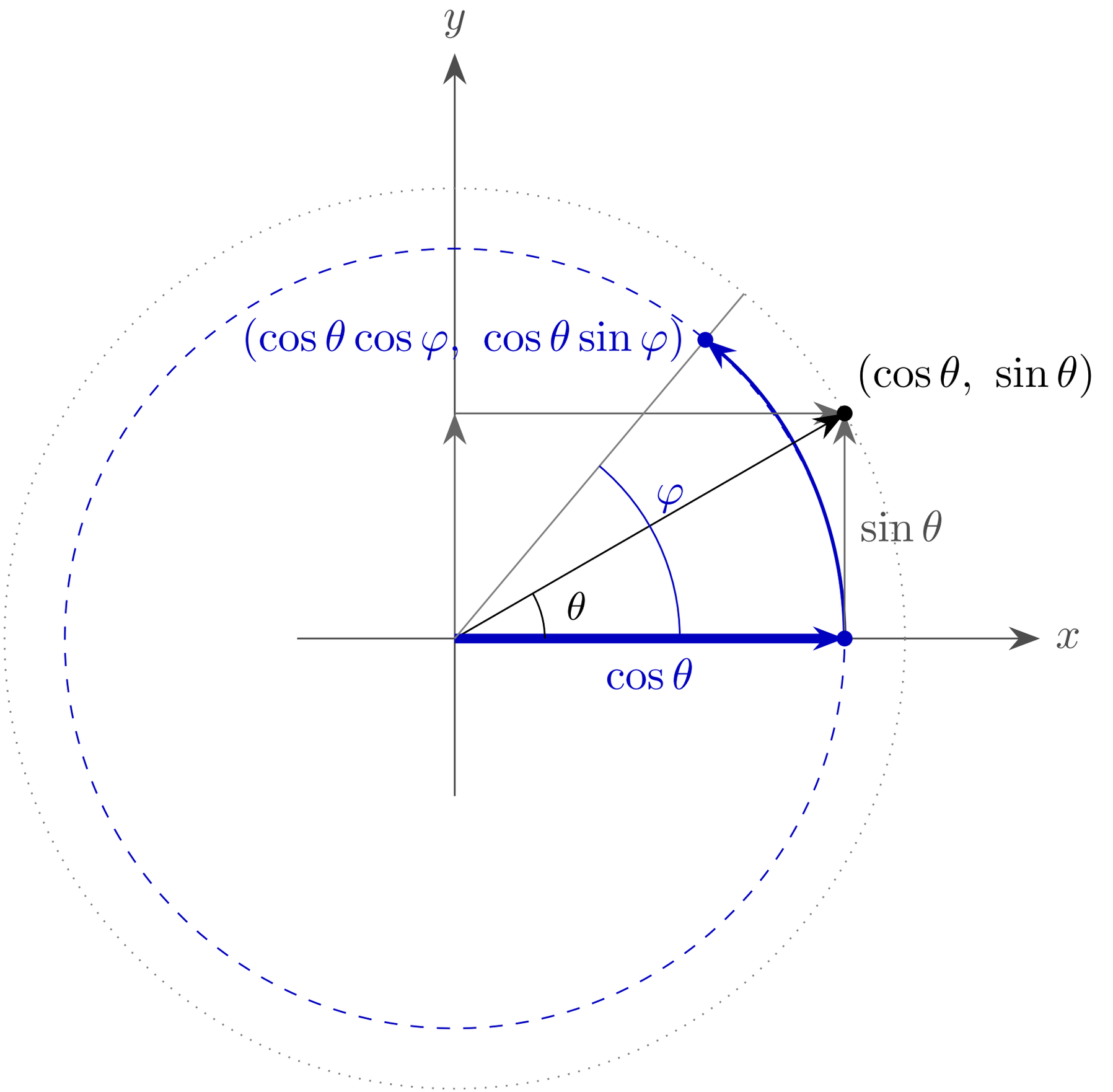
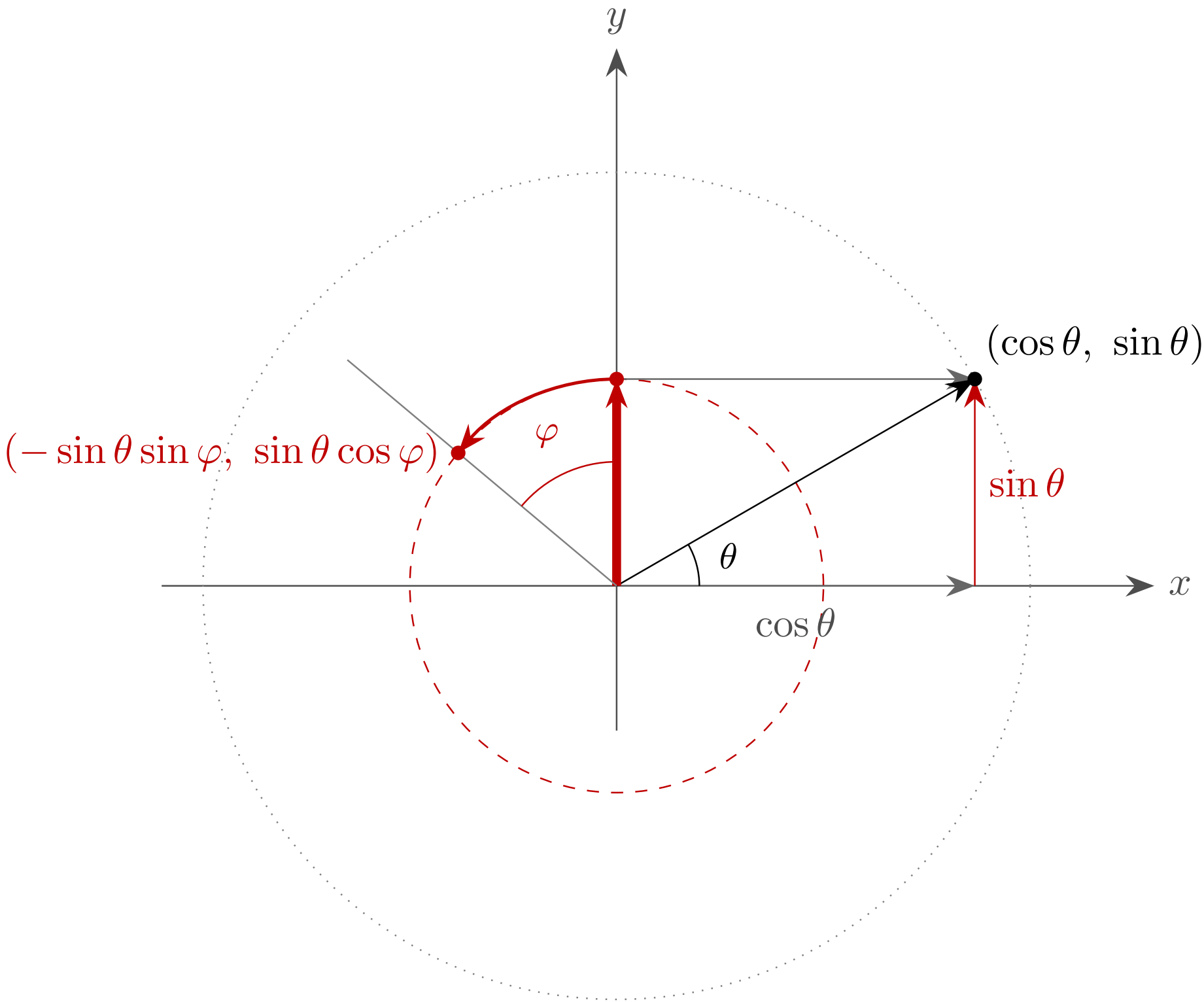
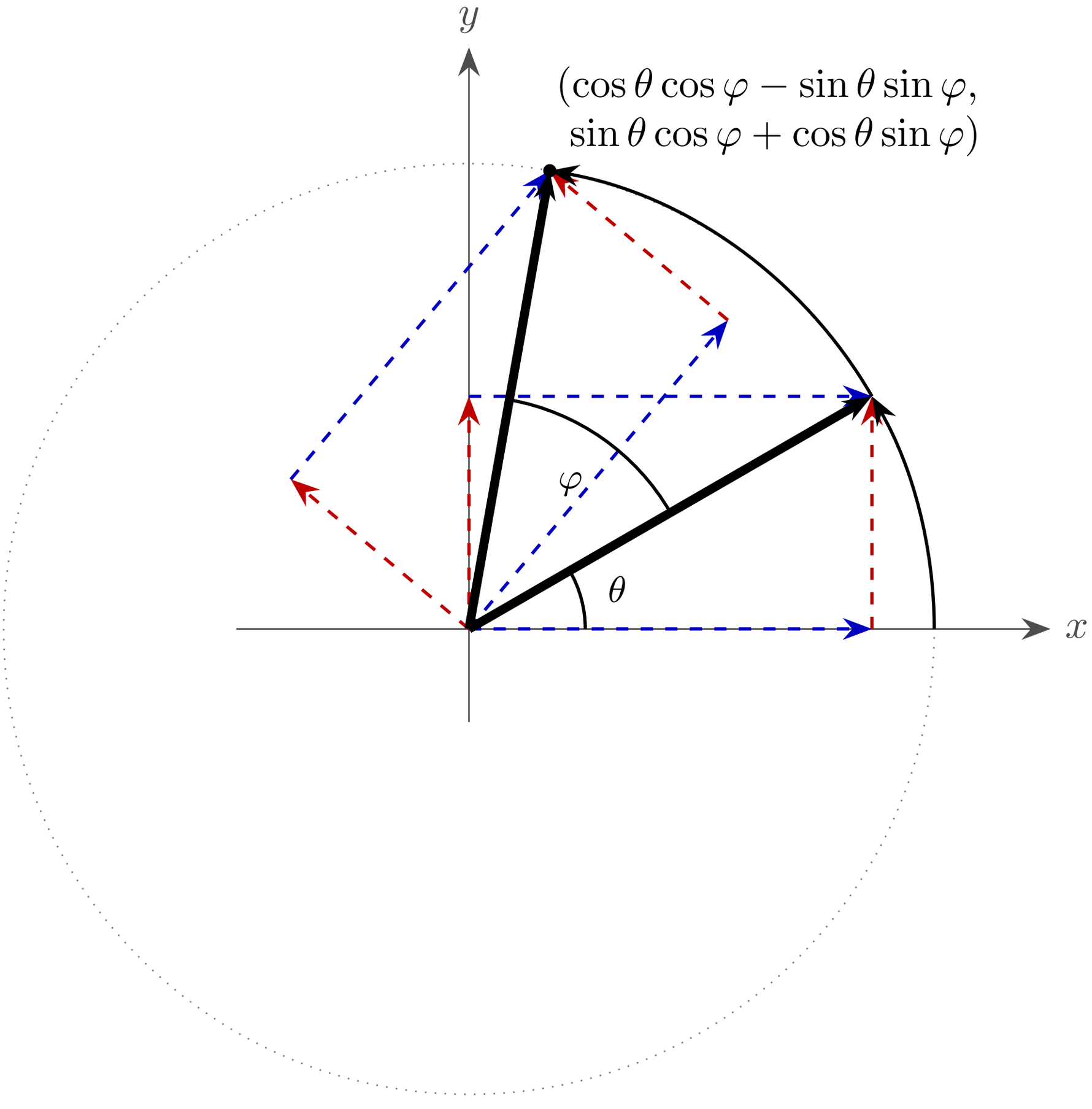

y
(0, 1)
(cos φ, sin φ)
(− sin φ, cos φ)
φ
φ
x
(1, 0)


y
$(\cos\theta\cos\varphi,\ \cos\theta\sin\varphi)$
$(\cos\theta,\ \sin\theta)$
φ
$\sin\theta$
θ
x
$\cos\theta$


y
$(\cos\theta,\ \sin\theta)$
$(-\sin\theta\sin\varphi,\ \sin\theta\cos\varphi)$
φ
$\sin\theta$
θ
x
$\cos\theta$


y
$(\cos\theta\cos\varphi - \sin\theta\sin\varphi,$
$\sin\theta\cos\varphi + \cos\theta\sin\varphi)$
φ
θ
x